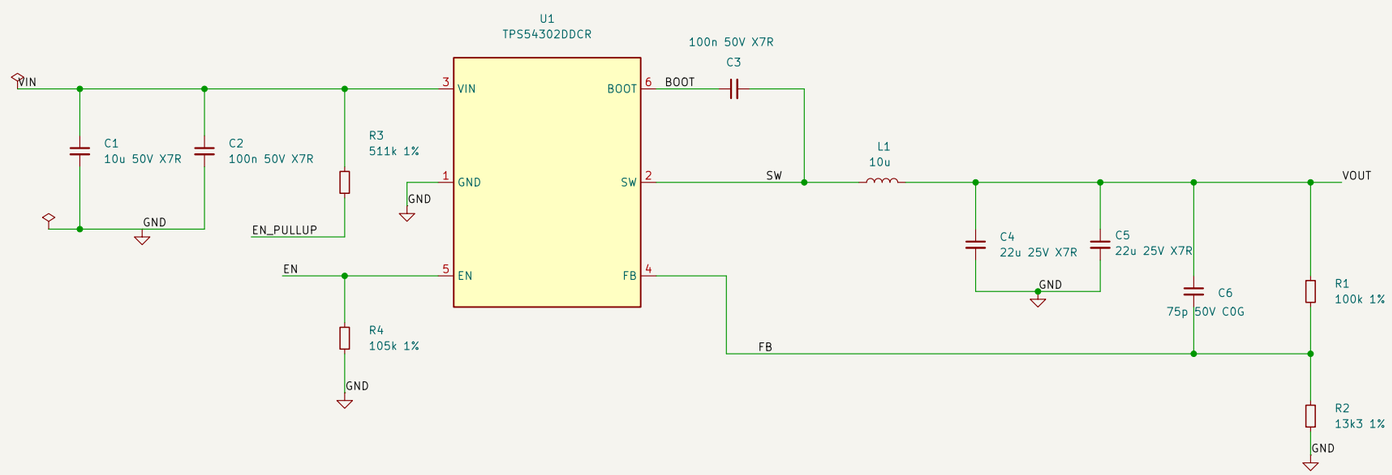
01 BATTERY TO 5 V | TI reference arrangement

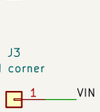


02 INPUT

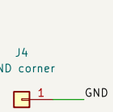
J1
VIN screw terminal



J3
VIN corner



J4
GND corner

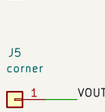


03 OUTPUT

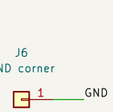
J2
5V screw terminal



J5
5V corner



J6
GND corner



04 ENABLE

J7
EN: 1–2 ON / 2–3 OFF



1–2: ON | 2–3: OFF

No cap: OFF

11–17 V input; 1 A normal target; 2–3 A brief peaks require bench verification.

Common GND. J1/J2: TOP. Headers J3–J7: BACK, pins down. 5 V assembly fitted.

05 5 V ASSEMBLY (FITTED)

REF	VALUE / PART	PACKAGE / SIDE
U1	TPS54302DDCR	TSOT-23-6
C1	10 uF, 50 V, X7R	1210
C2	100 nF, 50 V, X7R	0805
C3	100 nF, 50 V, X7R	0805
C4	22 uF, 25 V, X7R	1210
C5	22 uF, 25 V, X7R	1210
C6	75 pF, 50 V, C0G	0805
L1	10 uH, +/-20%	6 x 6 mm
R1	100 kOhm, 1%	0805
R2	13.3 kOhm, 1%	0805
R3	511 kOhm, 1%	0805
R4	105 kOhm, 1%	0805
J1, J2	XY127V-5.0-2P, 5 mm	THT / TOP
J3–J6	1x01 male, 2.54 mm	THT / BACK
J7	1x03 male, 2.54 mm	THT / BACK
J7 cap	1–2 ON / 2–3 OFF	2.54 mm

06 3.3 V ASSEMBLY (STARTING VALUES)

REF	VALUE / PART	PACKAGE / SIDE
U1	TPS54302DDCR	TSOT-23-6
C1	10 uF, 50 V, X7R	1210
C2	100 nF, 50 V, X7R	0805
C3	100 nF, 50 V, X7R	0805
C4	22 uF, 25 V, X7R	1210
C5	22 uF, 25 V, X7R	1210
C6	47 pF, 50 V, C0G	0805
L1	6.8 uH, +/-20%	6 x 6 mm
R1	100 kOhm, 1%	0805
R2	22.1 kOhm, 1%	0805
R3	511 kOhm, 1%	0805
R4	105 kOhm, 1%	0805
J1, J2	XY127V-5.0-2P, 5 mm	THT / TOP
J3–J6	1x01 male, 2.54 mm	THT / BACK
J7	1x03 male, 2.54 mm	THT / BACK
J7 cap	1–2 ON / 2–3 OFF	2.54 mm

3.3 V: TI TPS54302 Rev C, Table 7-2, p18.

Our R1/R2 = TI upper/lower divider; C6 = feed-forward.

Vnom = 0.596(1 + R1/R2): 5.077 V / 3.293 V.

3.3 V population needs its own stability / load checks.

Team Anant | AnantSAT-II | EM

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File: module_buck5V_TPS54302.kicad_sch

Title: Battery to 5 V – TPS54302

Size: A3	Date: 2026-10-17	**Rev: 0.6**
KiCad E.D.A. 10.0.6		Id: 1/1